Homework 2

● Graded

Student

Toby Hansen

Total Points

91 / 92 pts

Question 1

PO **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts Missing solution or totally wrong

(i) 5 points

– 1 pt minor mistakes in the table

– 2 pts major mistakes in the table

– 1 pt did not label the table with justifications

– 1 pt missing a few justifications or minor errors in the justification

– 2 pts missing too many justifications or major errors in the justifications

– 4 pts too much wrong

– 5 pts missing solution or too much wrong

– 2 pts did not fill out table

(ii) 5 points

– 2 pts weak or no justification

– 1 pt a small number of missing or incorrect Hasse Diagrams (has 8-9 or 11-12) or minor errors

– 3 pts too many incorrect or missing Hasse Diagrams (has <8 or >12) or major errors

– 1 pt Hasse diagrams contain transitive edges

– 5 pts no solution or very incorrect solution

– 4 pts minor justifications with incorrect number of partial orders and no hasse diagrams

– 3 pts did not draw Hasse diagrams

– 0.5 pts hasse diagram has correct number of edges but edges are not going up

Question 2

ER **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts Missing solution or totally wrong

(i) 5 points

– 1 pt minor mistakes in the table

– 2 pts major mistakes in the table

– 1 pt did not label the table with justifications

– 1 pt missing a few justifications or minor errors in the justification

– 2 pts missing too many justifications or major errors in the justifications

– 4 pts too much wrong

– 5 pts missing solution or too much wrong

(ii) 5 points

– 1 pt a small number of missing or incorrect partitions (off by 1)

– 3 pts too many missing or incorrect partitions (off by 2-4)

– 4 pts very incorrect solution (ie. off by >4)

– 5 pts no solution or extremely incorrect solution

– 1 pt Did not write partitions in the required alphabetized format

Question 3

P2.1.7 **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts missing or completely incorrect solution

(a) 5 points

– 5 pts part (a) missing or completely incorrect solution

– 1 pt correct idea but confusing explanation

– 3 pts correct basic idea but very confusing or incorrect explanation

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(b) 2.5 points

– 2.5 pts part (b) missing or completely incorrect solution

– 1.5 pts mildly incorrect description of when to give up

– 1 pt correct solution but suboptimal, (e.g. stop in $|X|$ steps)

(c) 2.5 points

– 2.5 pts incorrectly stated that a can sometimes be reached or did not state anything

– 1.5 pts correctly stated that a can sometimes not be reached but did not provide example or provided incorrect example

Question 4

P2.3.3 **9** / 10 pts

– **0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– **10 pts** missing or completely in correct solution

(a) 2.5 points

– **2.5 pts** solution to (a) missing or completely incorrect

– **2 pts** stated it was true and gave an explanation that had some correct parts

– **1 pt** did not give counterexample

(b) 2.5 points

– **2.5 pts** solution to (b) missing or completely incorrect

– **2 pts** stated it was true and gave an explanation that had some correct parts

– **1 pt** did not give counterexample

(c) 2.5 points

– **2.5 pts** solution to (c) missing or completely incorrect

– **2 pts** stated it was false and had (incorrect) counterexample

✔ – **1 pt** did not give sufficient explanation

3 The intution is certainly correct, but this is not really a proof - you should have explained how the existence of that element along with the given premise actually guarantees the existence of some element of B satisfying Q.

(d) 2.5 points

– **2.5 pts** solution to (d) missing or completely incorrect

– **2 pts** stated it was true and gave an explanation that had some correct parts

– **1 pt** did not give counterexample

– **1 pt** Multiple minor errors or general lack of explanation across all parts

1 67

2 6,7

4 If you were more wrong, I could have given you 6, 7 points, but alas.

5 6,7

Question 5

P2.3.8 **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts missing or completely in correct solution

(a) 3 points

– 3 pts solution to (a) missing or completely incorrect

– 1 pt partially correct solution but some errors

– 2 pts very badly incorrect

– 1 pt uses unallowed notation (ie. $\exists x, y, z$)

(b) 3 points

– 3 pts solution to (b) missing or completely incorrect

– 1 pt partially correct solution but some errors

– 2 pts very badly incorrect

– 1 pt uses unallowed notation (ie. $\exists x, y, z$)

(c) 4 points

– 4 pts solution to (c) missing or completely incorrect

– 2 pts partially correct solution but some errors

– 2 pts very badly incorrect

– 1 pt uses unallowed notation (ie. $\exists x, y, z$)

Question 6

P2.6.10 **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts missing or completely in correct solution

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translations (5 points)

– 1 pt Incorrect or missing translation of Premise I

– 0.5 pts Partially Incorrect translation of Premise I

– 1 pt Incorrect or missing translation of Premise II

– 0.5 pts Partially Incorrect translation of Premise II

– 1 pt Incorrect or missing translation of Premise III

– 0.5 pts Partially Incorrect translation of Premise III

– 1 pt Incorrect or missing translation of Premise IV

– 0.5 pts Partially Incorrect translation of Premise IV

– 1 pt missing or Incorrect translation of Premise V

– 0.5 pts partially Incorrect translation of Premise V

– 0.5 pts notation issues

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dog proof (5 points)

– 1 pt proof contains some minor errors

– 2 pts proof is generally OK but missing some formal details, e.g., not always explaining when Instantiation, Specialization, etc are used

– 3 pts proof contains some major errors

– 5 pts missing or totally incorrect proof

Question 7

P2.9.5 **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts missing or completely incorrect solution

(a) 2 points

– 2 pts solution to part (a) missing or totally incorrect

– 1.5 pts solution to part (a) contains major errors

– 0.5 pts solution to part (a) contains slight errors

– 1 pt did not list the injections

– 0.5 pts did not list all the 6 injections properly

(b) 2 points

– 2 pts solution to part (b) missing or totally incorrect

– 1.5 pts solution to part (b) contains major errors

– 0.5 pts solution to part (b) contains slight errors

– 1 pt did not list the surjections

– 0.5 pts did not list all the 6 surjections properly

(c) 3 points

– 3 pts solution to part (c) missing or totally incorrect

– 2 pts solution to part (c) contains major errors

– 1 pt solution to part (c) contains slight errors

– 1 pt did not discuss which could be written as $g \circ f$

– 0.5 pts did not correctly note that all the functions could be expressed as $g \circ f$

(d) 3 points

– 3 pts solution to part (d) missing or totally incorrect

– 2 pts solution to part (d) contains major errors

– 1 pt solution to part (d) contains slight errors

– 1 pt did not discuss which could be written as $f \circ g$

– 0.5 pts did not correctly note that only 21 of the functions could be expressed as $f \circ g$

– 0.5 pts did not explain why only 21 of the functions could be expressed as $f \circ g$

Question 8

P3.1.4 **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts missing or completely incorrect solution

(a) 2 points

– 2 pts has missing or totally incorrect solution with no work

– 1 pt has incorrect solution but with explanation showing where the error was

– 0.5 pts correct but did not explicitly write out the number

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(b) 3 points

– 3 pts has missing or totally incorrect solution with no work

– 1.5 pts has incorrect solution but with explanation showing where the error was

– 0.5 pts correct but did not explicitly write out the number

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(c) 5 points

– 5 pts missing or wrong solution for (c)

– 3 pts incorrect solution that has some correct parts

– 2 pts generally correct but description is too vauge or unclear (ie. did not mention taking the *product* of the *highest power* of every prime factor present in *either* number)

Question 9

EC: P2.1.10 **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts missing or completely incorrect solution

(a) 5 points

– 5 pts missing solution

– 3 pts description is not clear or correct enough to get full points

– 1 pt minor errors

(b) 5 points

– 5 pts missing solution

– 4 pts Did not fully describe $Q[][]$ or the $Q[]$ described could not provide a counterexample

– 1.5 pts Did not provide an Assignment that assigned every candidate to some job

– 1.5 pts Did not describe some ordering and choices made by the greedy algorithm on this input that has to stop before assigning all candidates to jobs

Question 10

EC: latex **2** / 2 pts

✔ **– 0 pts** used latex

– 2 pts not latex

Question assigned to the following page: 10

250 Homework #2 (Fall 2026)

Toby Hansen *

This assignment is due at 11:59 p.m. on Friday 9 October 2026, to be submitted on Gradescope. Late assignments (with penalty, unless excused) will be accepted until 11:59 p.m. on Saturday 10 October 2026, and not after. Please see the syllabus for more details on homework grading and late penalties.

When submitting to Gradescope, please do not forget to assign pages to problems.

All problems, with the exception of the first two and , originally come from the textbook. Some of them have been modified. In case of doubt, the version printed here is the one you should follow.

The instructions for this homework are the same as for HW1. Please read them there.

See the Canvas Homework web page for detailed instructions on how to format solutions. Solutions that do not follow those formatting instructions will be returned for rewriting and resubmission. In later assignments there will be points deducted for such formatting errors.

*Collaborated with Nobody.

Questions assigned to the following page: 1 and 10

Question 1 [10 pts]

Your solution should follow the format given on pages 17-19 of the relation study guide.

Let $A = \{a, b, c, d, e\}$. The relation $P \subseteq A \times A$ is a **partial order**.
It is known that

1. $(a, b), (b, c), (d, e), (d, b) \in P$ and
2. $(c, e), (e, b), (d, a), (b, d) \notin P$.

(i) In the table given below, label all pairs that must be in P with a "✓" and all pairs that cannot be in P with a "×." For pairs that are neither, leave their entry blank. We start you off by labelling the five known pairs with the appropriate "✓" or ×.

For full credit, you must justify your answers. Do that by numbering your justifications and, next to each ✓ or × in the table, write the number of the justification for that entry.

(a,a)		(b,a)		(c,a)		(d,a)	×2	(e,a)	
(a,b)	✓1	(b,b)		(c,b)		(d,b)	✓1	(e,b)	×2
(a,c)		(b,c)	✓1	(c,c)		(d,c)		(e,c)	
(a,d)		(b,d)	×2	(c,d)		(d,d)		(e,d)	
(a,e)		(b,e)		(c,e)	×2	(d,e)	✓1	(e,e)	

(ii) How many partial orders P' exist such that (a, b), (b, c), (d, e) and (d, b) are in P', and that satisfy that (c, e), (e, b), (d, a), and (b, d) are not in P'? Briefly explain how you know this.

Draw a separate Hasse diagram for each such partial order P'.
Note. You must draw these using the convention taught in class: an edge corresponding to relation (a, b) must go *up* from a to b.

Solution:

(i)

(a,a)	✓3	(b,a)	×5	(c,a)	×5	(d,a)	×2	(e,a)	×6
(a,b)	✓1	(b,b)	✓3	(c,b)	×5	(d,b)	✓1	(e,b)	×2
(a,c)	✓4	(b,c)	✓1	(c,c)	✓3	(d,c)	✓4	(e,c)	
(a,d)		(b,d)	×2	(c,d)	×5	(d,d)	✓3	(e,d)	×5
(a,e)		(b,e)		(c,e)	×2	(d,e)	✓1	(e,e)	✓3

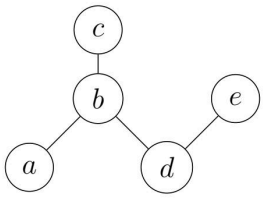
1. Its given that $(a, b), (b, c), (d, e), (d, b) \in P$
2. Its given that $(c, e), (e, b), (d, a), (b, d) \notin P$.
3. Since its a partial order, it must be reflexive; therefore every identity pair must be present: $\forall x \in A, (x, x) \in P$; $(a, a), (b, b), (c, c), (d, d), (e, e) \in P$

4. Since its a partial order, it must be transitive; therefore for any pairs in (1.) where the output is the same as another pair's input, the first pair's input must be paired with the second pair's output:
   a. Since $(a, b), (b, c) \in P$; $(a, c) \in P$
   b. Since $(d, b), (b, c) \in P$; $(d, c) \in P$
5. Since its a partial order, it must be antisymmetric; therefore every known pair from (1.) and (4.) CANNOT have a matching reverse pair (unless the elements are identical like identity pairs): $(b, a), (c, b), (e, d), (b, d), (c, a), (c, d) \notin P$
6. From transitivity, any pairs that would require a pair we've already ruled out to be present, cannot be in P; Since $(a, b) \in P$, and $(e, b) \notin P$; $(e, a) \notin P$

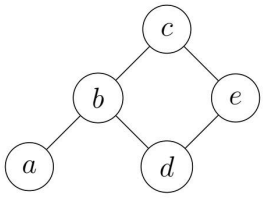
The entries for $(a, d), (a, e), (b, e), (e, c)$ are left blank because the given facts cannot be used to definitively derive whether those pairs *must* or *cannot* be in P.

(ii) There are 16 theoretical combinations of the four optional pairs (2^4), but only **10** of those 16 are actually possible with the given conditions.

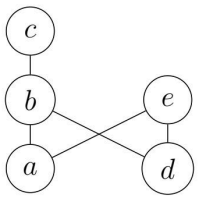
P01: $(a, d), (a, e), (b, e), (e, c) \notin P$
This is the minimal partial order with none of the four optional pairs included. Transitivity provides the (a, c) and (d, c) pairs in the graph.



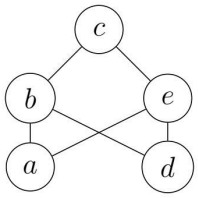
P02: $(a, d), (a, e), (b, e) \notin P$ and $(e, c) \in P$
This partial order includes 1 of the four optional pairs (e, c). None of the other optional pairs need to be included through transitivity. Transitivity provides the $(a, c), (d, c)$ pairs in the graph.



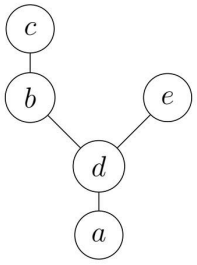
P03: $(a, d), (e, c), (b, e) \notin P$ and $(a, e) \in P$
This partial order includes 1 of the four optional pairs (a, e). None of the other optional pairs need to be included through transitivity. Transitivity provides the $(a, c), (d, c)$ pairs in the graph.



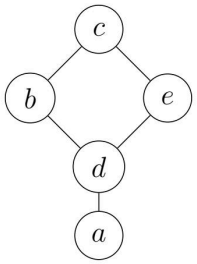
P04: $(a,d),(b,e) \notin P$ and $(a,e),(e,c) \in P$
This partial order includes the 2 optional pairs $(a,e),(e,c)$ that can be flipped independently of the other optional pairs (as seen in P03 and P02). Transitivity provides the $(a,c),(d,c)$ pairs in the graph.



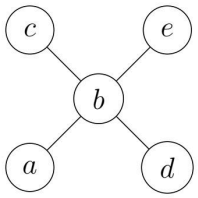
P05: $(b,e),(e,c) \notin P$ and $(a,d),(a,e) \in P$
Including (a,d) REQUIRES including (a,e) through transitivity, since (d,e) exists in all variations of P. Transitivity provides the $(a,c),(a,b),(a,e),(d,c)$ pairs in the graph.



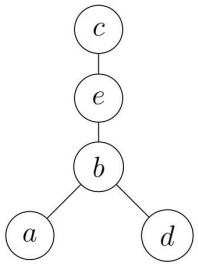
P06: $(b,e) \notin P$ and $(a,d),(a,e),(e,c) \in P$
This is the same as P04, just with the independent (e,c) included as well. Transitivity provides the $(a,c),(a,b),(a,e),(d,c)$ pairs in the graph.



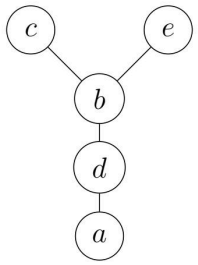
P07: $(a,d),(e,c) \notin P$ and $(b,e),(a,e) \in P$
Including (b,e) REQUIRES including (a,e) through transitivity, since (a,b) exists in all variations of P. Transitivity provides the $(a,c),(d,e),(a,e),(d,c)$ pairs in the graph.



P08: $(a,d) \notin P$ and $(b,e),(a,e),(e,c) \in P$
This is the same as P06, just with the independent (e,c) included as well. Transitivity provides the $(a,c),(d,e),(a,e),(d,c),(b,c)$ pairs in the graph.

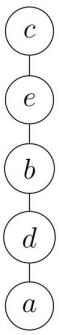


P09: $(e,c) \notin P$ and $(a,d),(b,e),(a,e) \in P$
This is the same as P06, but since (a,d) ALSO requires including (a,e), we can make a version with all 3. Transitivity provides the $(a,c),(a,b),(a,e),(d,c),(d,e)$ pairs in the graph.



P10: $(e,c),(a,d),(b,e),(a,e) \in P$
This is the same as P08, just with the independent (e,c) included as well. Transitivity provides the $(a,c),(a,b),(a,e),(d,c),(d,e),(d,c)$ pairs in the graph.

Questions assigned to the following page: 1 and 10



Questions assigned to the following page: 2 and 10

Question 2 [10 pts]

Your solution should follow the format given on pages 20-23 of the relation study guide.

Let $A = \{a, b, c, d, e, f, g\}$. The relation $Q \subseteq A \times A$ is an **equivalence relation**. It is known that

1. $(a, b), (c, d), (e, f) \in Q$, and
2. $(c, f), (g, e) \notin Q$.

(i) In the table given below, label all pairs that must be in Q with a "✓" and all pairs that cannot be in Q with a "×." For pairs that are neither, leave their entry blank. We start you off by labelling the five known pairs with the appropriate "✓" or ×.

For full credit, you must justify your answers. Do that by numbering your justifications and, next to each ✓ or × in the table, write the number of the justification for that entry.

(a,a)			(b,a)			(c,a)			(d,a)			(e,a)			(f,a)			(g,a)	
(a,b)	✓1		(b,b)			(c,b)			(d,b)			(e,b)			(f,b)			(g,b)	
(a,c)			(b,c)			(c,c)			(d,c)			(e,c)			(f,c)			(g,c)	
(a,d)			(b,d)			(c,d)	✓1		(d,d)			(e,d)			(f,d)			(g,d)	
(a,e)			(b,e)			(c,e)			(d,e)			(e,e)			(f,e)			(g,e)	×2
(a,f)			(b,f)			(c,f)	×2		(d,f)			(e,f)	✓1		(f,f)			(g,f)	
(a,g)			(b,g)			(c,g)			(d,g)			(e,g)			(f,g)			(g,g)	

(ii) (a) How many equivalence relations Q' exist that satisfy $(a, b), (c, d), (e, f) \in Q'$, and $(c, f), (g, e) \notin Q'$?

(b) Also, write down, in partition format, all of the equivalence relations that satisfy the 5 known pairs. It is not necessary to show how you derived them.

When writing down a partition, each of the subsets should be written in alphabetical order with each set being written in alphabetical order and the order of the sets being the alphabetical ordering of their 'first' element. For example, the equivalence relation that has g, c, e together, d, b together and a, f together should be written as $\Big\{\{a, f\},\ \{b, d\},\ \{c, e, g\}\Big\}$.

Solution:

(i)

(a,a)	✓3		(b,a)	✓5		(c,a)			(d,a)			(e,a)			(f,a)			(g,a)	
(a,b)	✓1		(b,b)	✓3		(c,b)			(d,b)			(e,b)			(f,b)			(g,b)	
(a,c)			(b,c)			(c,c)	✓3		(d,c)	✓5		(e,c)	×5		(f,c)	×5		(g,c)	
(a,d)			(b,d)			(c,d)	✓1		(d,d)	✓3		(e,d)	×5		(f,d)	×5		(g,d)	
(a,e)			(b,e)			(c,e)	×4		(d,e)	×4		(e,e)	✓3		(f,e)	✓5		(g,e)	×2
(a,f)			(b,f)			(c,f)	×2		(d,f)	×4		(e,f)	✓1		(f,f)	✓3		(g,f)	×6
(a,g)			(b,g)			(c,g)			(d,g)			(e,g)	×5		(f,g)	×6		(g,g)	✓3

Questions assigned to the following page: 2 and 10

1. Its given that $(a,b),(c,d),(e,f) \in Q$
2. Its given that $(c,f),(g,e) \notin Q$
3. Since its an equivalence relation, it must be reflexive; therefore every identity pair must be present: $\forall x \in A, (x,x) \in Q$; $(a,a),(b,b),(c,c),(d,d),(e,e),(f,f),(g,g) \in Q$
4. Since its an equivalence relation, it must be transitive; therefore any pairs that would require a pair we've already excluded to be included, cannot be in Q since they imply a pair that CANNOT be included:
   - Since $(c,d) \in Q$, and $(c,f) \notin Q$; $(d,f) \notin Q$
   - Since $(e,f) \in Q$, and $(c,f) \notin Q$; $(c,e) \notin Q$
   - Since $(c,d),(e,f) \in Q$, and $(c,f) \notin Q$; $(d,e) \notin Q$
5. Since its an equivalence relation, it must be symmetric; therefore every known pair from (1.) MUST have a matching reverse pair included, and every known pair from (2.) and (4.) CANNOT have a matching reverse pairs included: $(b,a),(d,c),(f,e) \in Q$, and $(f,c),(e,g),(f,d),(e,c),(e,d) \notin Q$
6. Just like in (4.), because of transitivity, since $(e,f) \in Q$, and $(e,g) \notin Q$; $(f,g) \notin Q$; Also, just like in (5.), because of symmetry, $(g,f) \notin Q$

The remaining entries are left blank because the given facts cannot be used to definitively derive whether those pairs *must* or *cannot* be in Q.

(ii) (a) From part (i), the initial equivalence classes are $\{a,b\},\{c,d\},\{e,f\},\{g\}$.
Since (c,f) is excluded, the classes $\{c,d\}$ and $\{e,f\}$ cannot be merged. Similarly, since (g,e) is excluded, the classes $\{e,f\}$ and $\{g\}$ cannot be merged. With the given conditions, exactly **7** possible combinations exist.

(b)
1. $\Big\{\{a,b\},\{c,d\},\{e,f\},\{g\}\Big\}$ (The initial relation)
2. $\Big\{\{a,b,c,d\},\{e,f\},\{g\}\Big\}$ (Merge a,b and c,d)
3. $\Big\{\{a,b,e,f\},\{c,d\},\{g\}\Big\}$ (Merge a,b and e,f)
4. $\Big\{\{a,b,g\},\{c,d\},\{e,f\}\Big\}$ (Merge a,b and g)
5. $\Big\{\{a,b\},\{c,d,g\},\{e,f\}\Big\}$ (Merge c,d and g)
6. $\Big\{\{a,b,c,d,g\},\{e,f\}\Big\}$ (Merge a,b and c,d and g)
7. $\Big\{\{a,b,e,f\},\{c,d,g\}\Big\}$ (Merge a,b and e,f; Merge c,d and g)

Questions assigned to the following page: 3 and 10

Question 3. P2.1.7 [10 pts]

Let X be a set of at least two people. We will ask each person in X to write down the name of a person in X other than themself. Let R be the relation $\{\langle a, b\rangle : a\text{'s chosen name is } b\}$. Suppose we start with a given person a, move to the person that a wrote down, then to the person that they wrote down, and so forth until we reach a given person b. Let S be the relation

$$S = \{\langle a, b\rangle : \text{Starting from } a, \text{ we eventually reach } b \text{ by this process}\}.$$

(a) If from a, we reach a again before finding b, explain why we will never reach b.

(b) If $\langle a, b\rangle$ is not in S, when can we give up on the process, i.e., be sure that $\langle a, b\rangle$ is not in S.

(c) If $\langle a, b\rangle$ is not in S, will we definitely reach a. i.e., is $\langle a, a\rangle$ in S? If the answer is yes, prove it. If the answer is no, give a counterexample in which $\langle a, b\rangle \notin S$ but $\langle a, a\rangle \notin S$.

Solution:

(a) If from a, we reach a again before finding b, we will never reach b. This is because if we circle back to a, it must be a cycle/loop since each person only writes down a single name (each input only has 1 output); and since a loop repeats infinitely, we would never get the chance to reach b, which lies OUTSIDE the loop.

(b) We can give up on the process and conclude that $\langle a, b\rangle \notin S$ if and ONLY if, we see a person's name more than once before finding b. This is because seeing a person's name more than once signifies an infinite loop, and if we hadn't found b by then, it'll never come up in the infinite loop that is made up of people we've already checked.

(c) If $\langle a, b\rangle$ is not in S, that does NOT necessarily mean that $\langle a, a\rangle$ is in S. Just because an infinite loop is reached, does not mean that it has to include all the way back as far as a; for example, if a wrote down x, x write down y, and y wrote down x, then even though you started with a and got caught in an infinite loop, the loop is just going back and forth between x and y.

Questions assigned to the following page: 4 and 10

Question 4. P2.3.3 [10 pts]

Let A and B be two types such that A is a proper subset of B ($A \subset B$, i.e., so that B contains all the elements of A plus at least one other element). Let P be a unary predicate on A, and let Q be a unary predicate on B. Suppose that $\forall a : \forall b : P(a) \rightarrow Q(b)$ is true.

Which of the following four statements is guaranteed to be true? For each statement, explain why it is always true or give an example where it is false. (Hint: Consider the case where A is empty.)

(a) $(\forall a : P(a)) \rightarrow (\forall b : Q(b))$

(b) $(\forall b : Q(b)) \rightarrow (\forall a : P(a))$

(c) $(\exists a : P(a)) \rightarrow (\exists b : Q(b))$

(d) $(\exists b : Q(b)) \rightarrow (\exists a : P(a))$

Solution:

In English, the given condition that is always true is: For every element of A and B, if the element of A satisfies P, then the element of B must satisfy Q

(a) FALSE
In English, if every element of A satisfies P, then every element of B satisfies Q.
Although it sounds nearly identical to the given condition, consider the empty set for A; P will always evaluate to TRUE since there are no counterexamples, meanwhile if even 1 value of B (which must have at least 1 value in a addition to the empty set) doesn't satisfy Q, then the implication is false. For example: P(x) is defined as "*x is not equal to 0*", Q(x) is defined as "*x is equal to 2*", A is the empty set, B is the set 1;

(b) FALSE
In English, if every element of B satisfies Q, then every element of A satisfies P.
According to the given condition, whether or not elements of B satisfy Q has no bearing on whether or not elements of A satisfy P. For example: P(x) is defined as "*x is less than 1*", Q(x) is defined as *x is greater than 5*, A is the set 6, B is the set 6,7.

(c) TRUE
In English, if at least one element of A satisfies P, then at least one element of B satisfies Q.
Since there needs to be at least one element in A for this proposition, the empty set trick won't work this time, if we do set A to the empty set it will just be false since there isn't at least one element. With the given condition, it is impossible for an element of A to satisfy P without an element of B satisfying Q, therefore this statement is guaranteed to be true.

Questions assigned to the following page: 4 and 10

(d) FALSE

In English, if at least one element of B satisfies Q, then at least one element of A satisfies P.

According to the given condition, whether or not elements of B satisfy Q has no bearing on whether or not elements of A satisfy P. For example: P(x) is defined as "*x is less than 1*", Q(x) is defined as *x is equal to 7*, A is the set 6, B is the set 6,7.

Questions assigned to the following page: 5 and 10

Question 5. Modified from P2.3.8 [10 pts]

Let X be a set and $R(x)$ a unary predicate on X.

(a) Write a quantified statement that says "there is exactly one element x in X such that $R(x)$ is true".

(b) Write a quantified statement that says "there are exactly two elements x in X such that $R(x)$ is true".

(c) Write a quantified statement that says "there are at least three elements x in X with $R(x)$ true, and at least three with $R(x)$ false"

Note: (a) is so common that there is shorthand macro for writing it. " $\exists! x : R(x)$" denotes that there is a unique $x \in X$ satisfying $R(x)$, which is what (a) is stating. This problem is asking you how to write this from scratch.

Your statements may use the binary relations $x = y$ and $x \neq y$).

Solution:

(a) $\exists a \in X : (R(a) \wedge (\forall c \in X : (c \neq a) \rightarrow \neg R(c)))$

(b) $\exists a, b \in X : (R(a) \wedge R(b) \wedge (a \neq b) \wedge (\forall c \in X : (c \neq a \wedge c \neq b) \rightarrow \neg R(c)))$

(c) $\exists a, b, c, d, e, f \in X : (R(a) \wedge R(b) \wedge R(c) \wedge \neg R(d) \wedge \neg R(e) \wedge \neg R(f) \wedge (a \neq b) \wedge (a \neq c) \wedge (b \neq c) \wedge (d \neq e) \wedge (d \neq f) \wedge (e \neq f))$

Question 6. P2.6.10 [10 pts]

One day, a set of five dogs met after their morning walks to compare their observations of birds. The set D of dogs consisted exactly of Arly (a), Cardie (c), Duncan (d), Mia (m), and Whistle (w).

The set S of bird species in question consisted exactly of Bluebird (B), Crow (C), Heron (H), Mallard (M), and Woodpecker (W).

The predicate $R(x, y)$ means "dog x observed a bird of species y".

(a) Translate the following statements into symbolic language.

I. There was a species that was observed by all the dogs.

II. Arly observed a Crow and no other species.

III. Every dog other than Arly observed at least two different species.

IV. Each dog, other than Arly and Cardie, observed all the species that Cardie observed and at least one that Cardie did not. (Note: Different dogs may have observed different species that Cardie did not.)

V. Duncan was the one and only dog to observe a Woodpecker, and Whistle was the one and only dog to observe a Bluebird.

(b) Use statements (I)-(V) to prove that some dog observed a Heron, i.e., $\exists x : R(x, H)$.

Your proof should provide all steps. In particular, it should explicitly state when Instantiation, Specialization, etc are used.

Solution:

(a) I. $\exists y \in S : \forall x \in D : R(x, y)$

II. $R(a, C) \wedge \neg R(a, B) \wedge \neg R(a, H) \wedge \neg R(a, M) \wedge \neg R(a, W)$

III. $\forall x \in D : (x \neq a \rightarrow (\exists y, z \in S : (y \neq z) \wedge R(x, y) \wedge R(x, z)))$

IV. $\forall x \in D : (x \neq a \wedge x \neq c) \rightarrow (\exists y \in S : (R(x, y) \wedge \neg R(c, y)) \wedge \forall z \in S : (R(c, z) \rightarrow R(x, z)))$

V. $R(d, W) \wedge R(w, B) \wedge (\forall x \in D : (x \neq d \rightarrow \neg R(x, W)) \wedge (x \neq w \rightarrow \neg R(x, B)))$

(b) We want to prove that $\exists x \in D : R(x, H)$.

1. Statement (I) says theres a species that every dog observed. By instantiation, this species can be s. $\forall x \in D : R(x, s)$
2. Statement (II) says that Arly observed only a Crow, and since theres a species that EVERY dog observed, the species MUST be Crow, as its the ONLY one Arly observed. By substitution, $s = C$ and $\forall x \in D : R(x, C)$

Questions assigned to the following page: 6 and 10

3. Statement (V) says that Duncan was the only dog to observe Woodpecker, and Whistle was the only dog to observe Bluebird. Statement (III) says that every dog other than Arly observed at least two different Species. Cardie is not Arly, Duncan, or Whistle, so Cardie MUST have seen either Heron or Mallard (or both) in addition to Crow (which everyone has seen, given by step 2): $R(c, H) \vee R(c, M)$
4. (i) Case 1 from step 3 supposes $R(c, H)$; Since $c \in D$, generalization gives $\exists x \in D : R(x, H)$
   (ii) Case 2 from step 3 supposes both $R(c, H) \wedge R(c, M)$; Since $c \in D$, generalization gives $\exists x \in D : R(x, H)$
   (iii) Case 3 from step 3 supposes $R(c, M) \wedge \neg R(c, H)$. By statement (IV), every dog other than Arly and Cardie observed at least one species that Cardie did not (Heron, Bluebird, and Woodpecker). Duncan could have just observed a Woodpecker in addition, and Whistle could've just observed a Bluebird in addition; but that still leaves Mia, who couldn't have observed a Bluebird or Woodpecker, since Whistle and Duncan respectively were the ONLY dogs to observe them; therefore the only remaining species Mia could have observed in addition to the specias Cardie observed is Heron: thus $R(m, H)$ must be true, and since $m \in D$, generalization gives $\exists x \in D : R(x, H)$
5. Proof by cases proved that in all 3 cases, $\exists x \in D : R(x, H)$ is true; therefore $\exists x \in D : R(x, H)$

Questions assigned to the following page: 7 and 10

Question 7. P2.9.5 [10 pts]

Note: Be aware that, in the textbook, this problem spans two pages. Also, the solution to this one is long.

Let $A = \{v, w\}$ and $B = \{x, y, z\}$ be sets of characters.

(a) List all the possible functions from A to B. Determine which are injections.

(b) List all the possible functions from B to A. Determine which are surjections.

(c) How many possible functions are there from A to A? How many can be expressed as $g \circ f$, where f is a function from A to B and g is a function from B to A?

(d) How many possible functions are there from B to B? How many can be expressed as $f \circ g$, where g and f are functions as above?

Solution:

(a) A function is an injection when the distinct inputs have distinct outputs (when $f(v) \neq f(w)$).

1. $\{(v, x), (w, x)\}$ – NOT injection
2. $\{(v, x), (w, y)\}$ – injection
3. $\{(v, x), (w, z)\}$ – injection
4. $\{(v, y), (w, x)\}$ – injection
5. $\{(v, y), (w, y)\}$ – NOT injection
6. $\{(v, y), (w, z)\}$ – injection
7. $\{(v, z), (w, x)\}$ – injection
8. $\{(v, z), (w, y)\}$ – injection
9. $\{(v, z), (w, z)\}$ – NOT injection

(b) A function is a surjection if every element of the set the function is being applied *TO* is present in the outputs (both v and w appear as outputs).

1. $\{(x, v), (y, v), (z, v)\}$ – NOT surjection
2. $\{(x, v), (y, v), (z, w)\}$ – surjection
3. $\{(x, v), (y, w), (z, v)\}$ – surjection
4. $\{(x, v), (y, w), (z, w)\}$ – surjection
5. $\{(x, w), (y, v), (z, v)\}$ – surjection
6. $\{(x, w), (y, v), (z, w)\}$ – surjection

Questions assigned to the following page: 7 and 10

7. $\{(x, w), (y, w), (z, v)\}$ – surjection
8. $\{(x, w), (y, w), (z, w)\}$ – NOT surjection

(c) There are 4 possible functions from A to A (2^2), and all 4 can be expressed as $g \circ f$, where f is a function from A to B and g is a function from B to A; this is because f and g can both produce 2 outputs, exactly what ALL A to A functions need.

(d) There are 27 possible functions from B to B (3^3), but only 21 can be expressed as $f \circ g$, where f is a function from A to B and g is a function from B to A; this is because f and g can both only produce 2 outputs, so you have to exclude the 6 functions from B to B that have 3 distinct outputs.

Questions assigned to the following page: 8 and 10

Question 8. P3.1.4 [10 pts]

The **least common multiple** of two naturals x and y is the *smallest* natural that both x and y divide. For example, $lcm(8, 12) = 24$ because 8 and 12 each divide 24, and there is no smaller natural that both 8 and 12 divide.

(a) Find the least common multiple of 60 and 339.
Write it out as a number and not, e.g., as its prime factorization.

(b) Find the least common multiple of $2^3 \cdot 3^2 \cdot 5^4$ and $2^2 \cdot 3^4 \cdot 5^3$.
Write it out as a number and not, e.g., as its prime factorization

(c) Describe a general method to find the least common multiple of two naturals, given their factorization into primes (and assuming that the factorization exists and is unique). Write your solution in terms of $p_i(m)$ and $p_i(n)$ as described below.

Notation: Let p_i be the i'th prime. For natural m, let $p_i(m)$ be the exponent of p_i in the prime factorization of m, i.e., the largest power in p_i that divides m. So $m = \prod_{i=1}^{\infty} p_i^{p_i(m)}$. Example: $p_1 = 2$, $p_2 = 3$ and $p_3 = 5$. Since $20 = 2^2 \cdot 5^1$, $p_1(20) = 2$, $p_2(20) = 0$, $p_3(20) = 1$ and, $\forall i > 3$, $p_i(20) = 0$.

Solution:

(a) $60 = 2^2 \cdot 3 \cdot 5$ and $339 = 3 \cdot 113$
Therefore, $2^2 \cdot 3 \cdot 5 \cdot 113 = 6780$

(b) $2^3 \cdot 3^4 \cdot 5^4 = 405000$

(c) $lcm(m, n) = \prod_{i=1}^{\infty} p_i^{\max(p_i(m), p_i(n))}$
For each prime, choose the larger exponent, then multiply the prime powers together.

Questions assigned to the following page: 9 and 10

Extra Credit. P2.1.10 in the book. [10 pts]

Let C be a set of n candidates and J be a set of n jobs. Define the relation Q to be

$$Q = \{\langle c, j\rangle \ : \text{ candidate } c \text{ is qualified for job } j\}$$

Define the relation A to be

$$A = \{\langle c, j\rangle \ : \text{ candidate } c \text{ is assigned to job } j\}.$$

(a) We say that candidate c is "possible" for job j if they are qualified for j, they are not assigned to any other job, and no one else is assigned to j. Informally describe how you would determine, given boolean arrays for the relations Q and A, whether c is possible for j.
Note: This problem is essentially asking you to design an algorithm to verify *whether a given assignment A of people to jobs is possible.*

(b) One way to assign candidates to jobs would be to start with an empty assignment and repeatedly find a possible candidate for a job and assign them to it, until there are no possible candidates left. This **greedy algorithm** will not necessarily assign the greatest possible number of candidates to jobs. Find an example where there is a possible assignment of qualified candidates for all the jobs, but there is a sequence of assignments of possible candidates that stops, with no further assignments possible, while some jobs are still unfilled.
Note: This is asking you to design just one *counterexample. The counterexample could be very small, e.g., only 2 or 3 people and jobs.*

Solution:

(a) (1.) Check whether $Q(c, j)$ is true, meaning that candidate c is qualified for job j. If it is false, then c is not possible for j; Otherwise...

(2.) Check whether $A(c, k)$ is false where k is all jobs other than j, meaning that candidate c has no other jobs. If even one relation A is true and shows that candidate c is assigned another job, then c is not possible for j; Otherwise... candidate c is possible for j.

(b) Consider two candidates $c1, c2$ and two jobs $j1, j2$ with $Q = (c1, j1), (c1, j2), (c2, j1)$ ($c1$ is qualified for both jobs while $c2$ is only qualified for $j1$). Going in sequence order, $c1$ would get assigned $j1$ since they are qualified with no other assignment, $c1$ would NOT get assigned $j2$ since they already have another assignment, $c2$ would NOT get assigned $j1$ since it is already assigned, $c2$ would NOT get assigned $j2$ since they are not qualified. Thus, even though $c2$ could be assigned $j1$, and $c1$ could be assigned $j2$, the greedy algorithm did not see it happen.